

Atlantic Sea Scallop Management History and Current Fishery Overview

Research Track Peer Review
Woods Hole, MA
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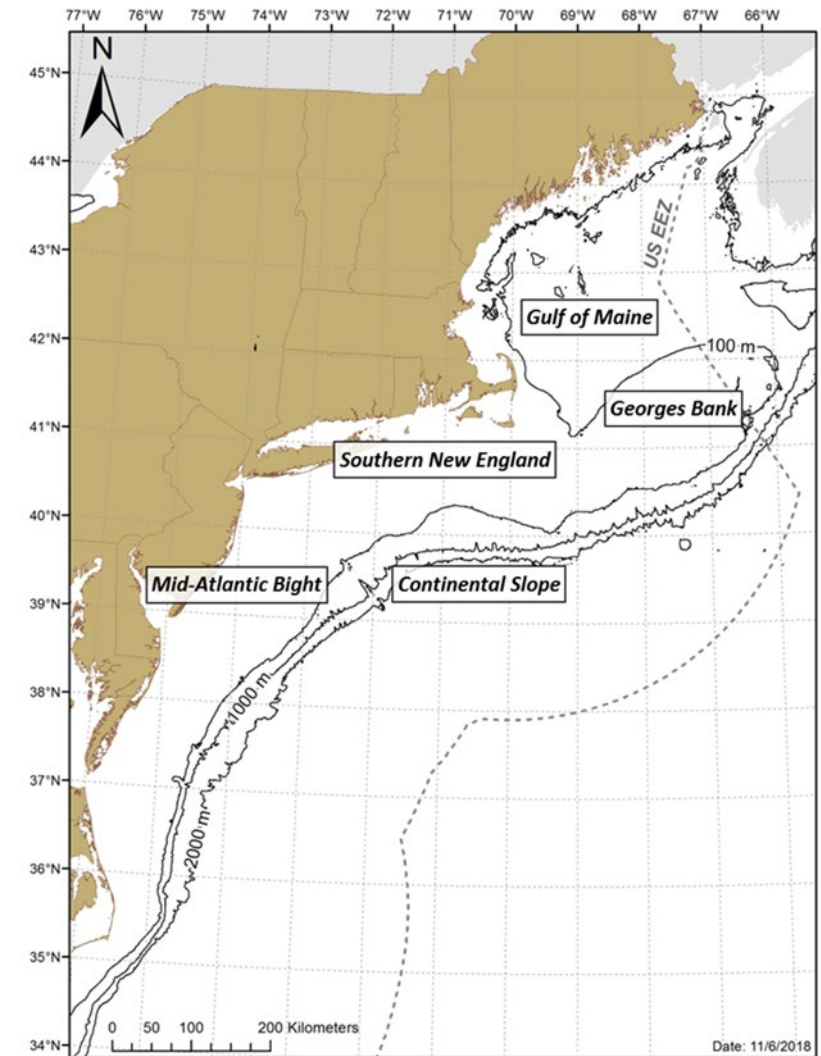


New England
Fishery Management Council



Species Overview

- Life History and Distribution
 - Bivalve mollusk from Cape Hatteras to Newfoundland, Canada.
 - Primarily in depths of 30-100 meters.
 - Patchy distribution with higher concentrations typically on hard sand and gravel substrates.
 - Filter feeders that are mature at age 2, but much more productive after age 4.
 - Larvae are planktonic for 1-2 months then settle to bottom.
 - Scallop meat weights affected by depth and latitude. In general, meat weight decreases with depth, and MA scallops do not grow as large as GB.



Fishery Overview

- Eastern North American continental shelf from Virginia to the Gulf of Maine – primarily harvested in the Mid-Atlantic Bight (MAB) and Georges Bank (GB).
- ~500 active vessels, single or paired dredge gear.
- 4" minimum ring size to select for larger scallops.
- Management measures consist of limited entry, effort controls (days-at-sea), gear and crew regulations, and rotational area management.
- Primary landing port: New Bedford, MA.





Federal Management Actions

- 1982 – The Atlantic Sea Scallop Fishery Management Plan is implemented.
- 1994 – NOAA Fisheries closes areas on Georges Bank and Nantucket shoals to groundfish and scallop fishing.
- 1994 - Moratorium on scallop permits implemented; limits on "days-at-sea" to reduce fishing effort and a limit on crew size are initiated; increase in dredge ring size from 3.0 to 3.5 inches; open access "general category" scallop permits created to allow small-scale scalloping and incidental catch of scallops in other fisheries (Amendment 4).
- 1997 – Scallops are declared overfished.
- 2004 - Area rotation management program implemented (rotating open and closed areas to maximize scallop yield); dredge ring size is further increased to 4 inches; minimum mesh size for mesh on top of the dredge gear set at 10 inches to reduce flounder bycatch (Amendment 10).
- 2006 - Chain mats required in sea scallop dredges used in the Mid-Atlantic, May through November, to prevent sea turtles from entering gear.
- 2008 - The limited access general category (LAGC) scallop fishery is created and there is no longer an open access scallop permit (Amendment 11).
- 2011 - Annual catch limit and accountability measure (AM) requirements of the Magnuson-Stevens Fishery Conservation and Management Act are implemented (Amendment 15).
- 2018 – Revisions to habitat closures re-open scallop grounds on inshore Georges Bank and south of Nantucket, spurring an increase in fishery landings.
- 2022 – Amendment 21 establishes a new allocation sharing arrangement to support research, monitoring, and a directed fishery with trips limits for LAGC NGOM and IFQ vessels below a biomass threshold in NGOM management area. All federally permitted scallop vessels to fish in the NGOM when/if biomass reaches higher levels; Increase the LAGC IFQ possession limit in open rotational areas (access areas) to 800 pounds per trip.

Permit Categories

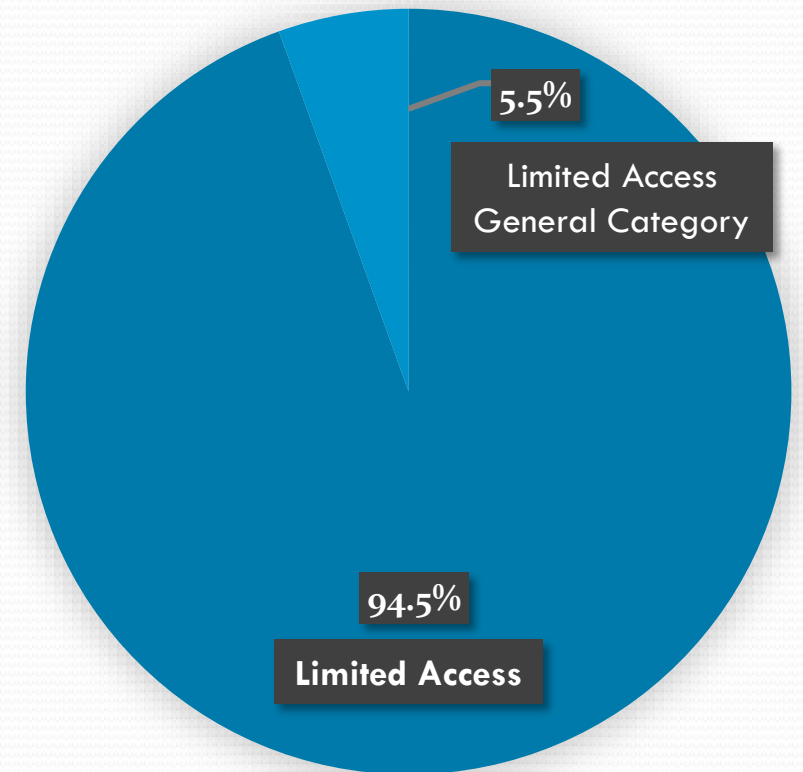
- Limited Access

- Harvest Limit: 94.5% of Annual Projected Landings
- Allocation level: Vessel level - **Days at sea** and **access area trips** are allocated at the vessel level
 - 8 vessel categories: Full/Part-time, Occasional, Full/Part-time small dredge, Full/Part-time/Occasional trawl net

- Limited Access General Category

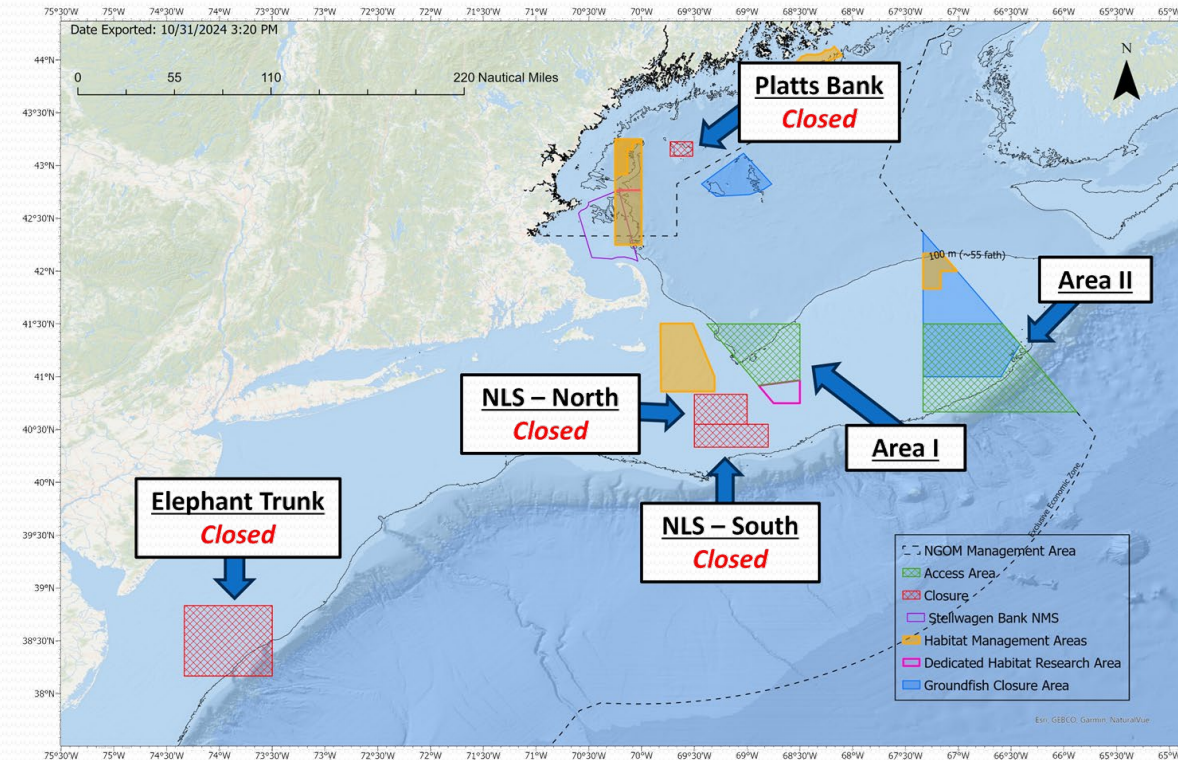
- *Individual Fishing Quota (IFQ)*
 - Harvest Limit: 5.5% of APL, after set-asides and incidental catch removed
 - Allocation level: Vessel level - IFQ (in pounds) and set number of trips to each access area at a fleet level
- *Northern Gulf of Maine (NGOM)*
 - Harvest Limit: Up to TAC for management area, not linked to APL estimate
 - Allocation level: Fishermen may harvest in area until LAGC fleet reaches Total Allowable Catch (TAC).
 - Only for Northern Gulf of Maine
- *Incidental Catch*
 - Harvest Limit: Deducted from APL before allocating to LA and LAGC IFQ
 - Allocation level: Allows fishermen to harvest up to 40lbs shucked or 5 bu in-shell scallops per trip

% APL Allocated To Each Permit Type



Rotational Management

- Rotating closures around areas of recruitment to protect small scallops, improve yield per recruit
- Directs fishing effort to areas with higher catch rates and larger scallops
 - Improved control of fishing mortality
 - Reduced reliance on effort controls
 - Decreased area swept
- Largely successful, but challenges have included:
 - Anomalous growth rates
 - Concentrated fishing effort in high-density areas
 - Increased natural mortality

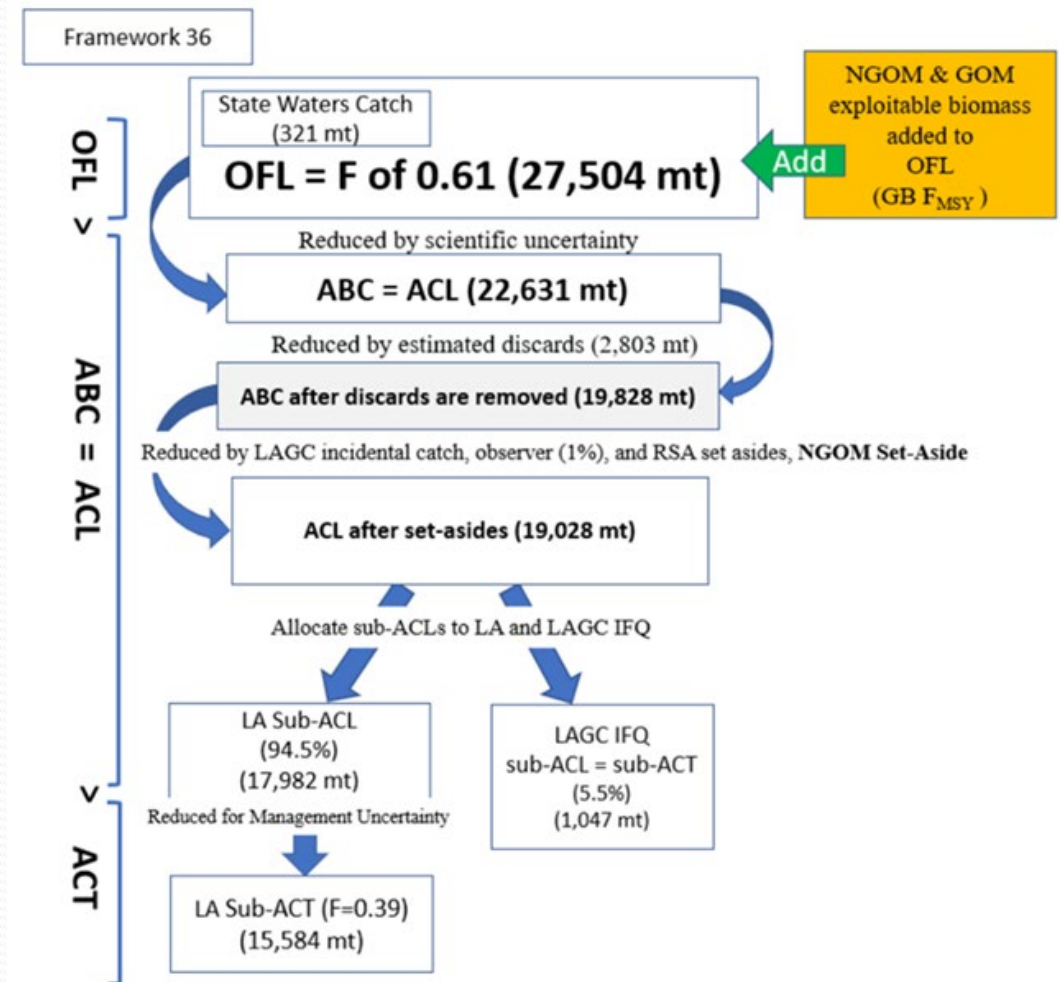


ABC Control Rule

OFL and ABC

- OFL - equivalent to the catch associated with F_{MSY} ($F=0.61$)
- ABC control rule – catch set at 75% probability of remaining below the F associated the OFL ($F=0.45$). Risk is evaluated in terms the probability of overfishing compared to the fraction loss of yield.

	Definition in Scallop FMP	SARC 50 (2010)	SARC 59 (2014)	SARC 65 (2018)	2020 Management Track
OFL	F_{MSY}	$F=0.38$	$F=0.48$	$F=0.64$	$F=0.61$
ABC=ACL	75% probability of remaining below the OFL	$F=0.32$	$F=0.38$	$F=0.51$	$F=0.45$
B_{MSY}	B_{TARGET}	125,358 mt	96,480 mt	116,766 mt	102,657 mt
$\frac{1}{2} B_{MSY}$	$B_{THRESHOLD}$	62,679 mt	48,240 mt	58,383 mt	51,329 mt
MSY		24,975 mt	23,798 mt	46,531 mt	32,079 mt
Overfished?	$B < B_{THRESHOLD}$	No	No	No	No
Overfishing?	$F < F_{THRESHOLD}=F_{MSY}$	No	No	No	No

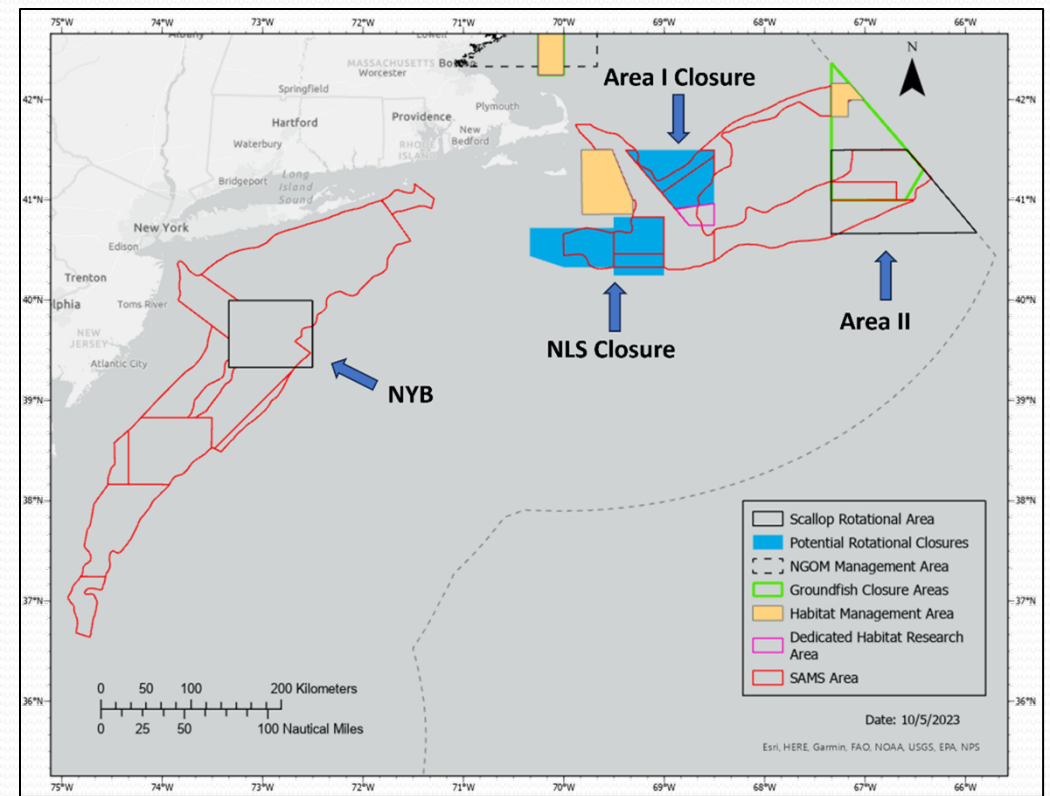
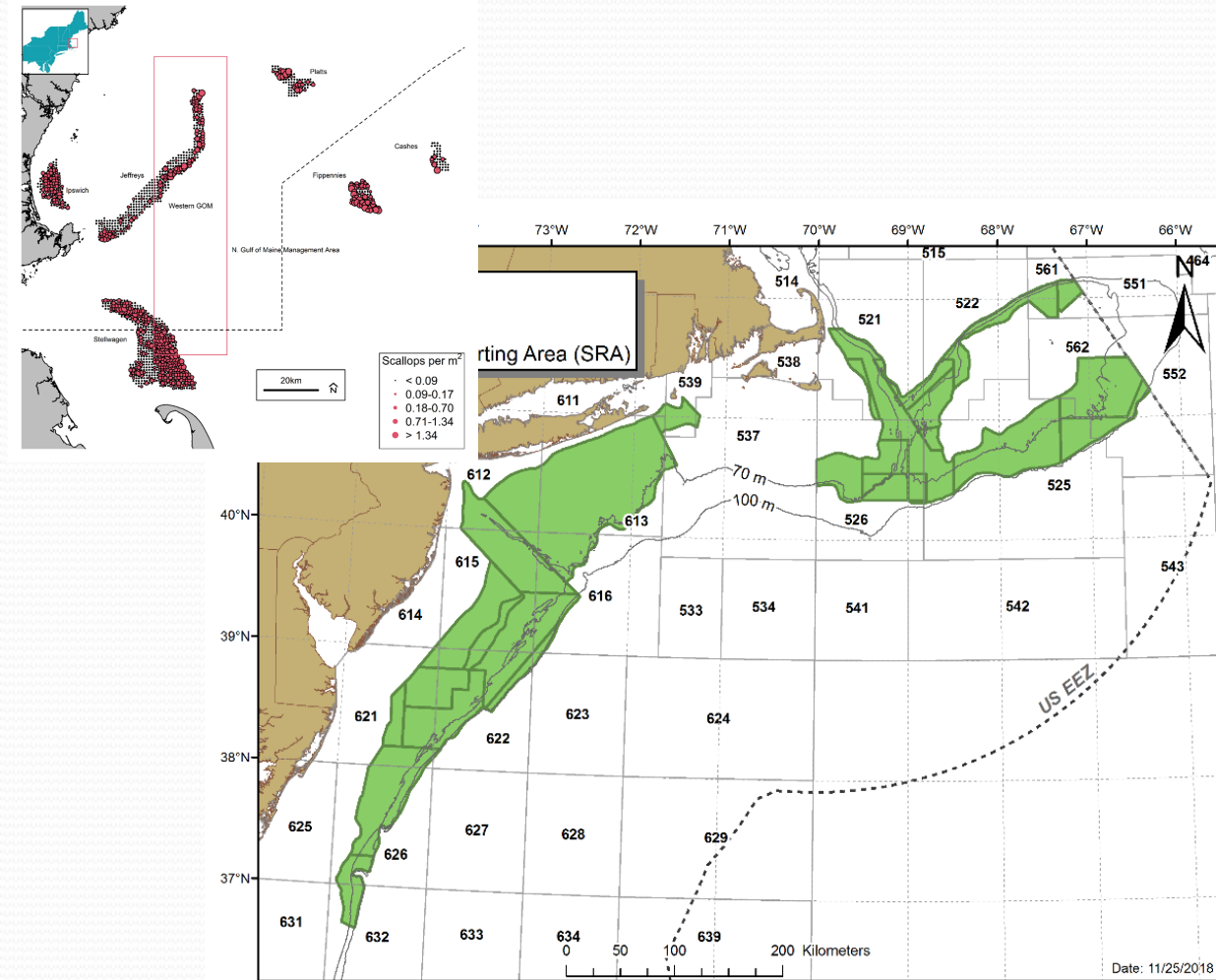


Overfishing Limit and Acceptable Biological Catch

(legal limits set using all survey data)

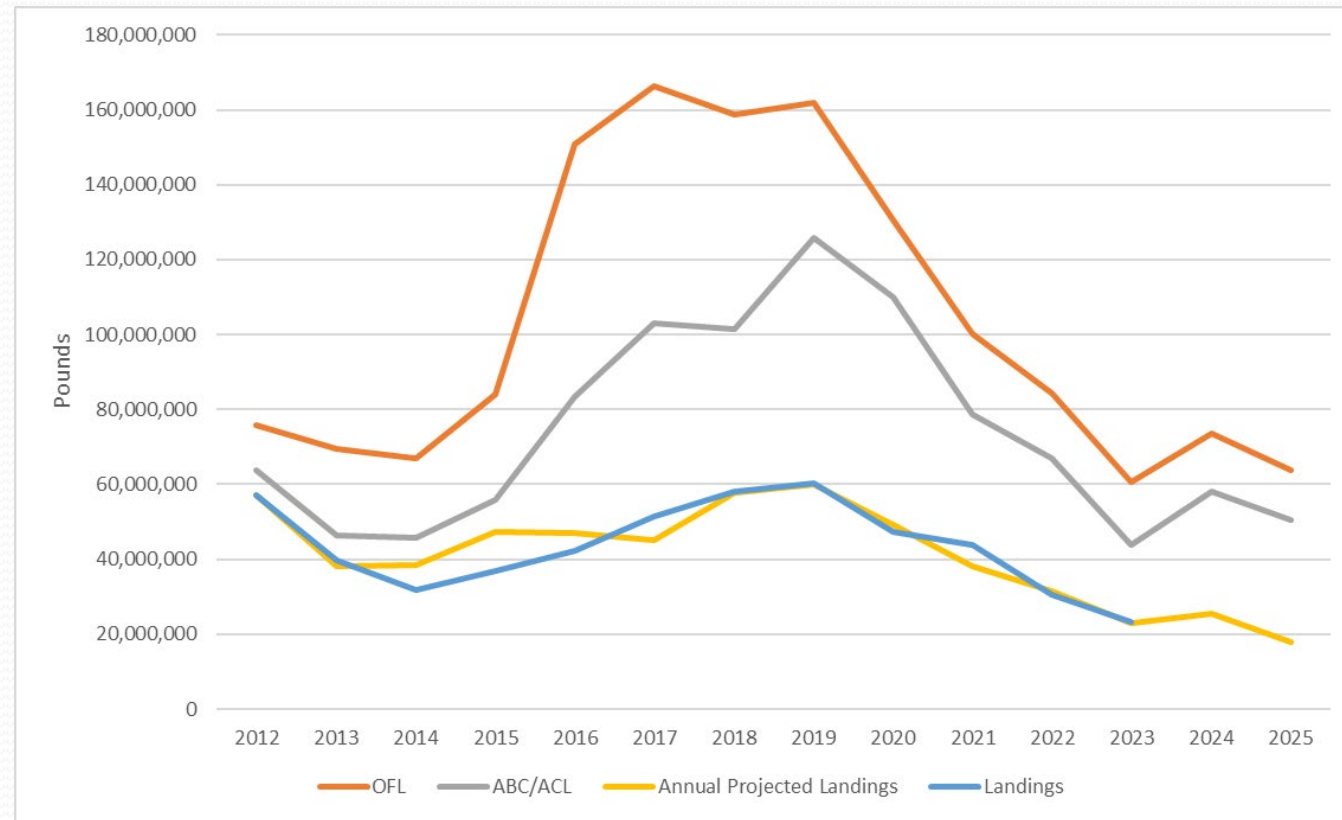
Annual Projected Landings

(anticipated landings with rotational closures and openings)



OFL, ABC, Projected Landings, Landings

Figure 1 - Overfishing limit, Acceptable Biological Catch/Annual Catch Limit, Annual Projected Landings, and Landings for fishing years 2012 - 2025.



- 2012 forward.
- Time period captures exceptional 2012 (GB) and 2013 (MA) year classes.
- Recent landings peak with reconfiguration of habitat closed areas.
- Steady decline in landings since 2019.

Fishing Effort, Terminal Year of Assessment (2023)

- VMS data. 30-minute pings. Speed filter of 2 – 5.5 kt applied. Minimum
- Majority of fishing effort on Georges Bank and in the Great South Channel.
- Effort in the Gulf of Maine concentrated around Stellwagen Bank.

